

Data Communication and Networking (ECC-311)

Data Communication -

- Sharing information or data.
- This sharing can be local or remote.
- Local communication → face to face.
- Remote communication → takes place over a distance.

Definition

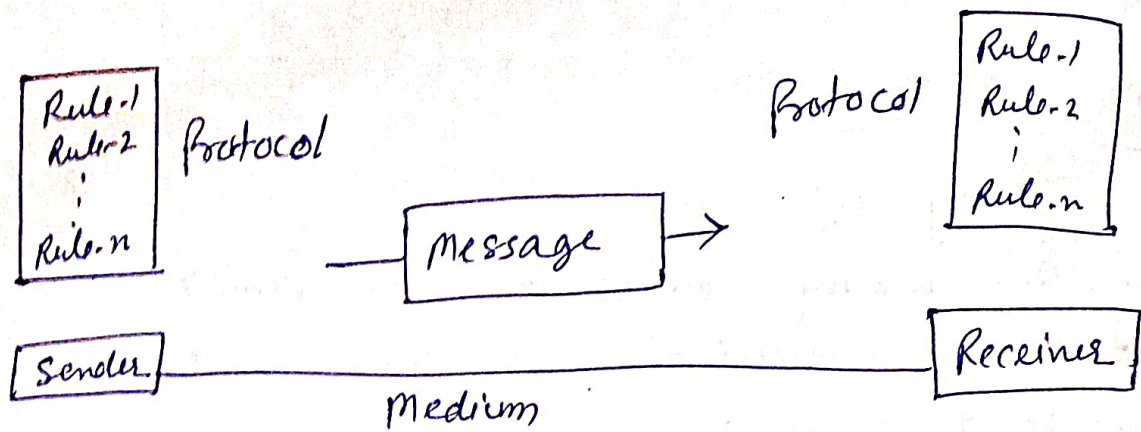
Data Communication is the exchange of data b/w two devices via some form of transmission medium such as a wire cable.

- The effectiveness of the data comm. s/s depends on four fundamental characteristics -

- (1) - Delivery - system must deliver data to the correct destination.
- (2) - Accuracy - must deliver the data accurately.
- (3) - Timeliness - must deliver data in timely manner.
- (4) - Jitter - Data delivered late are useless.
↓
↳ This time of delivery is called "real-time transmission".

- It refers to the variation in the packet arrival time.
- It is the uneven delay in the delivery of audio or video packets.

Components - five components.



- 1- Message - The message is the information (data) to be communicated.
eg - text, numbers, pictures, audio, video.
- 2- Sender - That sends the data message.
eg - computer, telephone, video camera etc.
- 3- Receiver - That receives the message.
eg - computer, workstation, telephone, television etc.
- 4- Transmission medium - physical path by which a message travel from sender to receiver. eg. Twisted pair wire, coaxial cable, fibre optic cable, and radio waves.
- 5- Protocol - It is a set of rules that govern data communication. It represents the agreement b/w communication devices. w/o a protocol, Two devices may be connected but not able to communicate.

Message

Text → It is represented as a bit pattern, a sequence of bits (0's and 1's).

- Today, the prevalent coding system is 'Unicode' which is 32-bits to represent a symbol or character.

Numbers → Represented by bit patterns.

- Unicode is not used to represent numbers.

Images → Also represented by a bit pattern.

- In its simplest form, an image is composed of a matrix of pixels (picture elements), where each pixel is small dot.
- The number of pixels depends on the resolution.
- Each pixel is assigned a bit pattern. The size and the value of pattern depends on the image.

eg. Black & white image - (a chessboard)

1-bit pattern is enough to represent a pixel.

⇒ If image is not pure white/black then increase the size of bit pattern to include the gray scale. eg. few levels of gray scale.

- 00 - Black pixel
- 01 - dark gray pixel
- 10 - light " "
- 11 - white pixel.

17 color image -

Date

One method is RGB.

- so called because each color is made up of a combination of three primary colors - Red, Green, Blue.
- The intensity of each color is measured and a bit pattern is assigned to it.
- Another method is called YCM, in which a color is made up of combination of three other primary colors: Yellow, cyan and magenta.

Audio - recording or broadcasting of sound ~~no~~ or music.

- Audio is different from text, number or images.
- It is continuous, not discrete.

Video - refers to the recording or broadcasting of a picture or movie. Video can either be produced as a continuous entity, or it can be a combination of images, each a discrete entity.

Data Flow

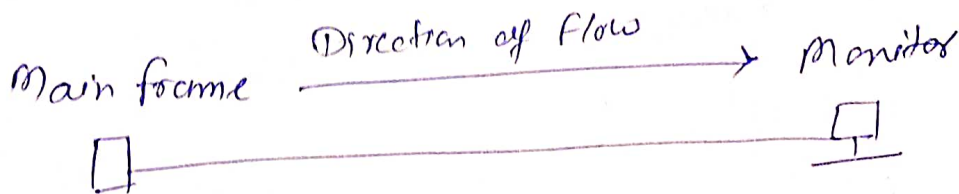
(3)

- Communication b/w Two devices can be -

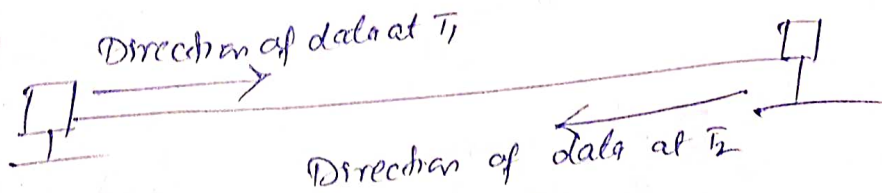
- Simplex, → unidirectional (One can transmit and other only receives)
- Half Duplex → Each station can transmit and receive, but not at the same time.
- Full Duplex



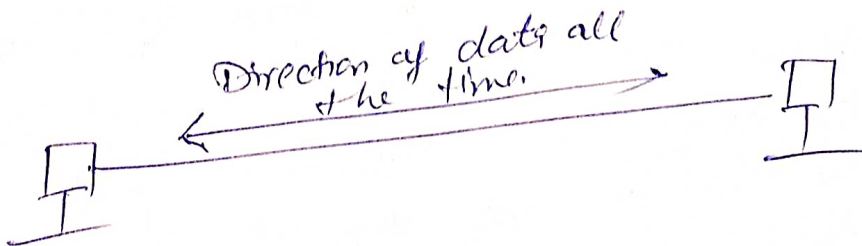
Both stations can transmit and receive simultaneously.



(A) - Simplex method



(B) - Half Duplex



(C) - Full Duplex.

Networks -

(4)

A network is the interconnection of a set of devices capable of communication.

- A device can be host, such as large computer, desktop, laptop, workstations, cellular phone, or security system. Some connecting devices such as -
- Router → Connect ~~two~~ network to other network.
- Switch → that connects device together.
- Modem → (modulator - demodulator) that changes the form of data.

Network Criteria →

- (1) - Performance - measured in transit time and response time.
- (2) - Reliability - measured by the frequency of failure, the time it takes a link to recover from a failure, and the n/w robustness in a catastrophe (disaster).
- (3) - Security → Network security includes protecting data from unauthorized access, protecting data from damage and development, and implementing policies and procedures for recovery from breaches and data losses.

Physical Structures -

Type of Connection -

- A n/w is two or more devices connected through links.
- There are two possible types of connections -
 - (1) - Point to point → Dedicated link b/w two devices.
 - (2) - Multi point → (multidrop). more than two devices share a link. → capacity of channel is shared ~~ex~~ either spatially or temporally.

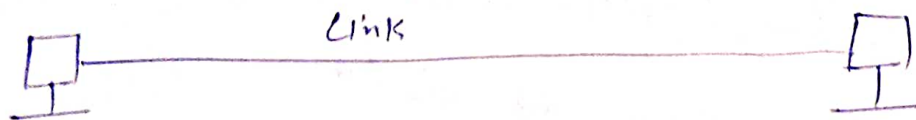


fig- Point to Point

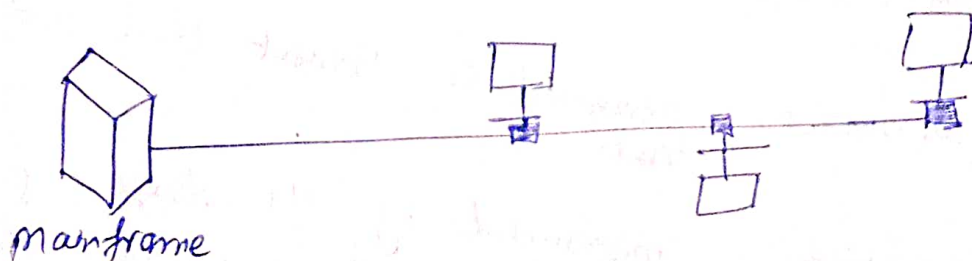


fig- multipoint

- If several devices can use the link simultaneously, it is a spatially shared connection.
- If users must take turns, it is a 'timeshared' connection.

Physical Topology -

(6)

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- The term physical topology refers to the way in which a n/w is laid out physically.
- The topology of n/w is the geometric representation of the ~~relationship~~ relationship of all the links and linking devices (usually called nodes) to one another.
- There are four basic topologies -
 - (1) - Mesh Topology
 - (2) - Star "
 - (3) - Bus "
 - (4) - Ring "

Mesh Topology -

- Every device has a dedicated point to point link to every other device.
- For fully connected mesh n/w with n - nodes $n(n-1)$ physical links.

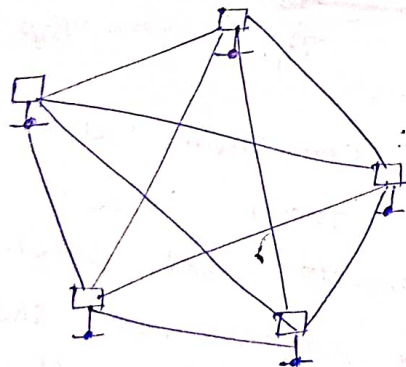


Fig - Mesh Topology
(Five devices)

- To accommodate that many links, every device on the n/w must have $(n-1)$ input/output ports to be connected to the other $(n-1)$ stations.

Star Topology -

- Each device has a dedicated point to point link only to a central controller, usually called a hub.
- The devices are not directly linked to one another.

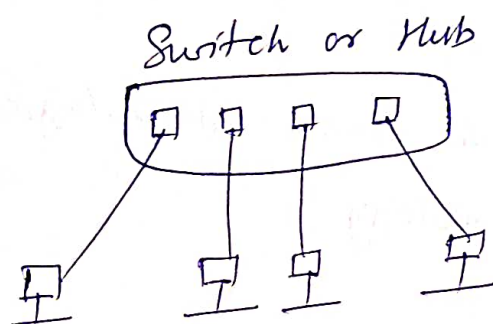
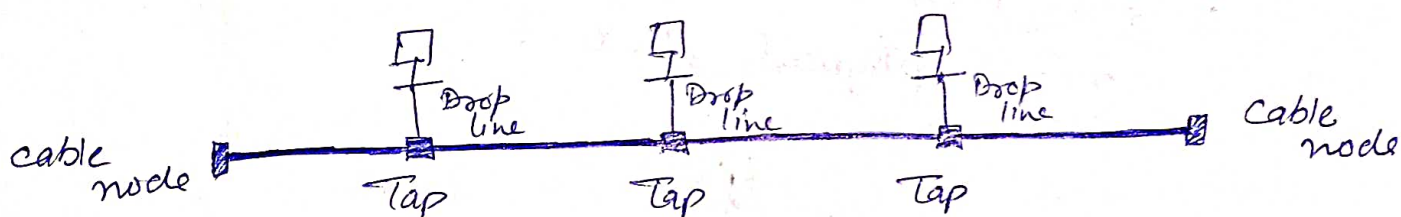


Fig - Star Topology

Bus Topology -

- Bus topology is multipoint.
- One long cable acts as a backbone to link all the devices in a network.



- Drop line is a connection running b/w the device and the main cable.
- Tap is a connector that either splices into the main cable or punctures the sheathing of a cable to create a contact with the metallic core.

Ring Topology.

(6)

- Each device has a dedicated point to point connection with only the two devices on either side of it.
- A signal is passed along the ring in one direction from device to device, until it reaches its destination.
- Each device in the ring incorporates a repeater, which generates the bits and passes them along.

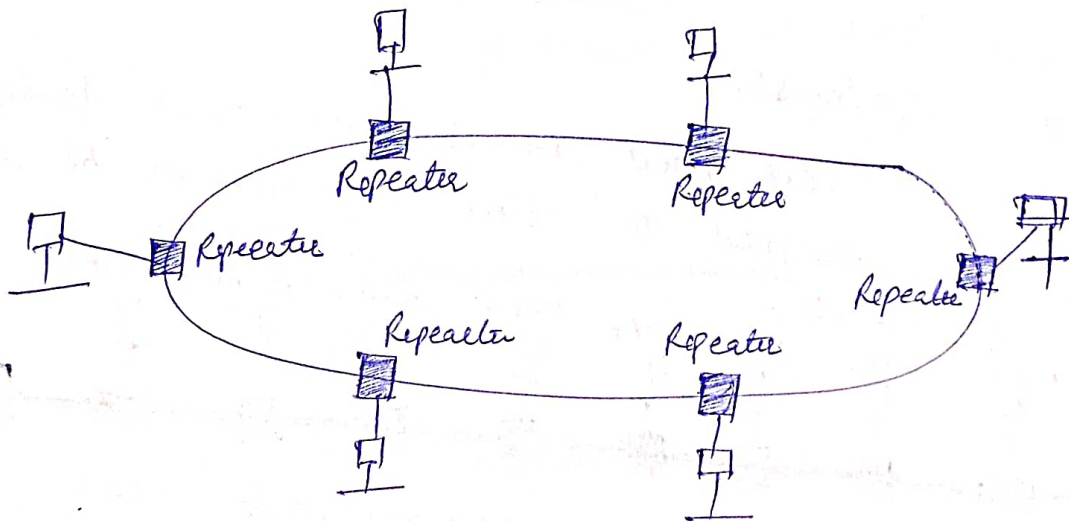


Fig- Ring Topology

Network Types ..

(I)- Local Area Network (LAN) -

- LAN is usually privately owned and connects some hosts in a single office, building, or campus.
- Each host in a LAN has an identifier, which is an address that uniquely defines the host in the LAN.
- A packet sent by a host to another host carries both the source and destination host's addresses.
- When LANs were used in isolation (rare today) they were designed to allow resources to be shared b/w the hosts.

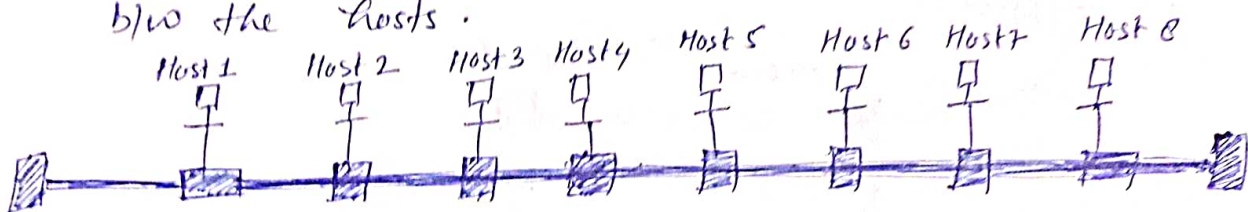


Fig - LAN with a common cable (Past)

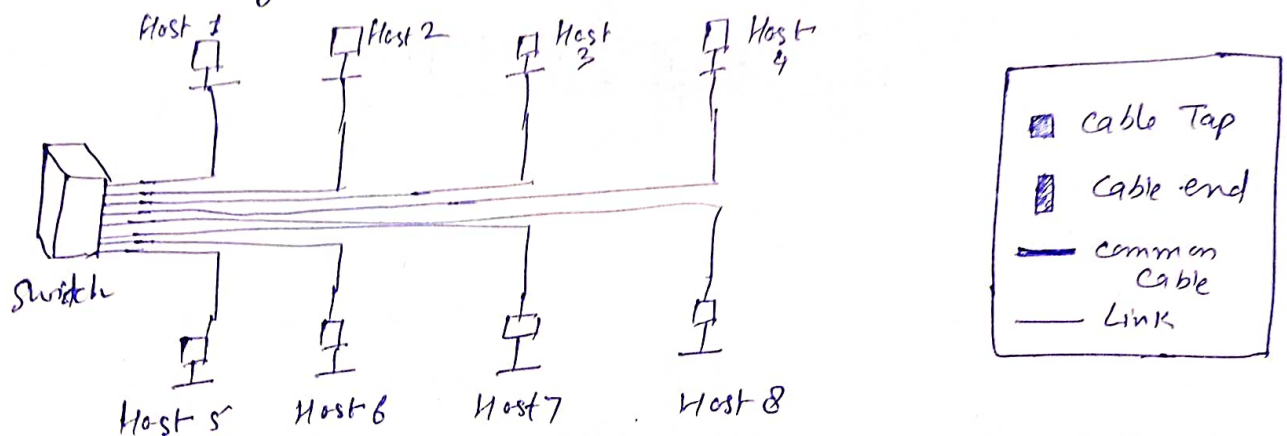


Fig - LAN with a switch (today).

1- Wide Area Network (WAN)

(7)

- Capable of communication at wider level.

Differences in b/w LAN & WAN

* A LAN is normally limited in size, spanning an office, a building, or a campus; a WAN has a wider geographical span, spanning a town, a state, a country, or even the world.

* A LAN interconnects hosts, a WAN interconnects connecting devices such as switches, routers, or modems.

- Two examples of WAN -

- ⇒ point to point WAN - connect two devices through (cable or air)
- ⇒ Switched WAN - with more than two ends
- Backbone of the global comm. n/w today.

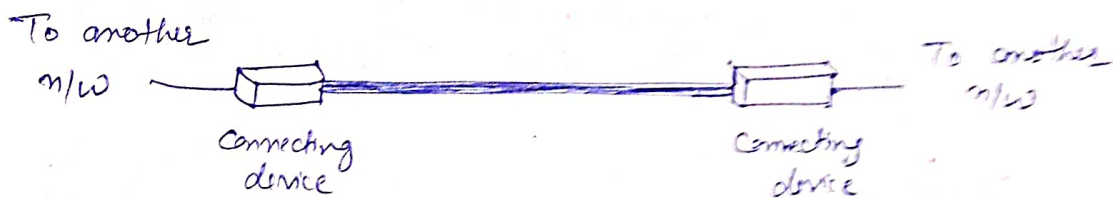


Fig. point to point WAN

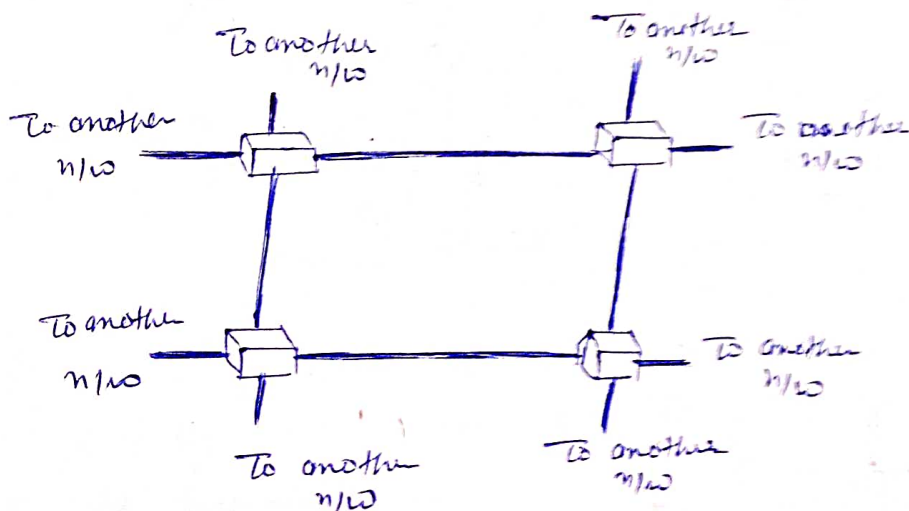


Fig. A switched WAN

Internetwork

(lowercase i) → private intern

Internet
to
Change

Internet network

Today, it is very rare to see LAN ~~and~~ WAN in isolation, they are connected to one another. When two or more n/w are connected, they make an internetwork, or internet.
(small caps)

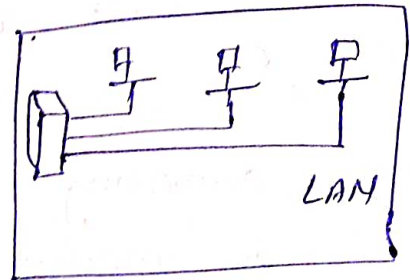
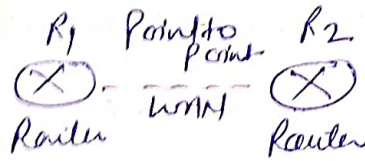
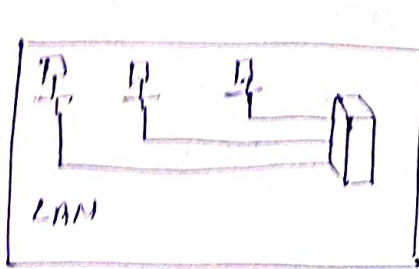


Fig - An internetwork made of Two LANs and one point to point WAN

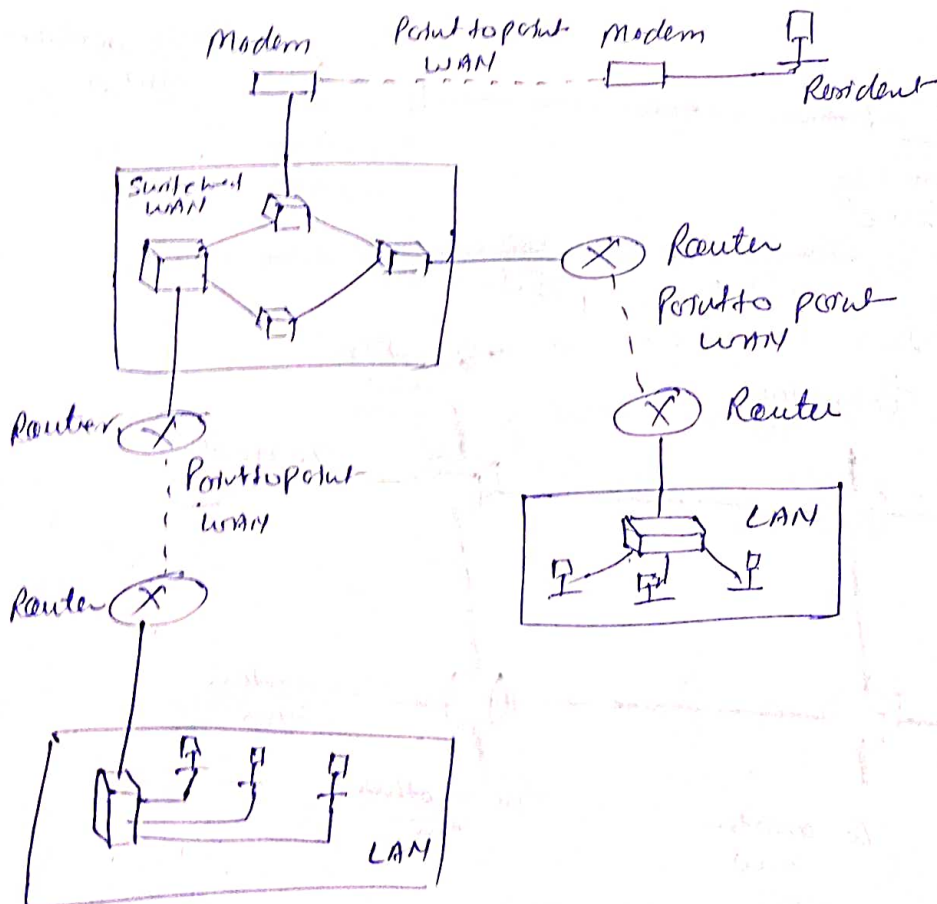


Fig - A heterogeneous internetwork made of few WANs and Two LANs.

Internet - (uppercase. I)
(Large caps)

Internet $\rightarrow$ (i) $\rightarrow$ Two or more n/w's that can communicate with each other.

Internet (I) $\rightarrow$ Composed of Thousands of interconnected n/w's.

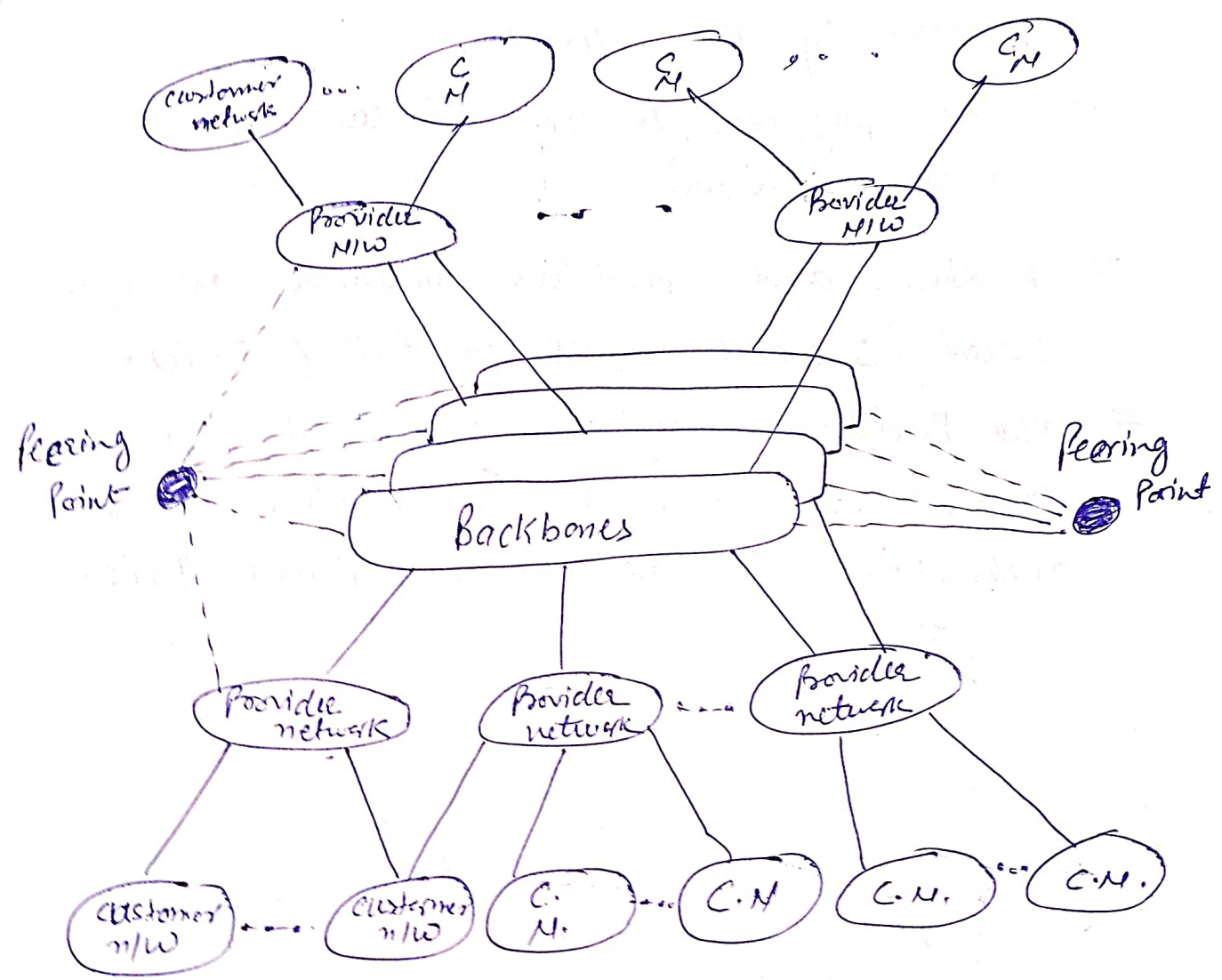


Fig - The Internet Today

- $\rightarrow$ At the Top level, the backbones are ~~large~~ large n/w's own by some communication companies.
- $\rightarrow$ The backbone n/w's are connected through some complex switching systems, called "Peering Points".

→ At the second level - There are smaller networks, called provider networks, that use the services of the backbones for a fee.

→ The Provider n/w are connected to backbones and sometimes to other provider networks.

→ The customer n/w are n/w at the edge of the Internet that actually use the services provided by the Internet.

→ They pay fees to the Provider n/w for receiving services.

Backbones and Providers networks are also called "Internet Service Providers" (ISPs).

The Backbones are often referred to as international ISPs; the Provider n/w are often referred to as national or regional ISPs.

Protocol layering -

(10)

(9)

- When the communication is complex, we may need to divide the task b/w different layers, in which case we need a protocol at each layer, or Protocol layering.

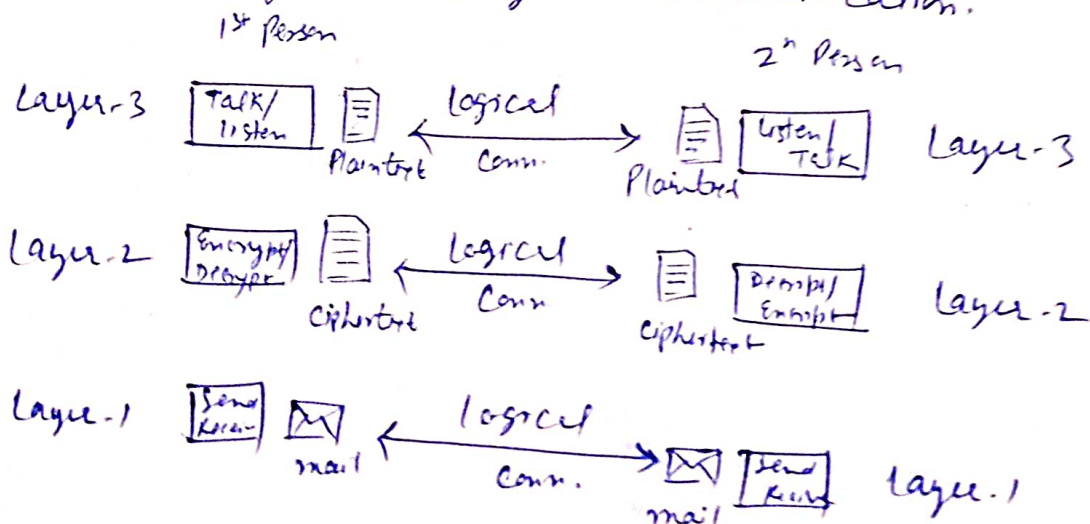
Principle of Protocol layering -

First Principle - If we want bidirectional communication, we need to make each layer so that it is able to perform two opposite tasks, one in each direction.

Second Principle - The Two objects under each layer at both sites should be identical.

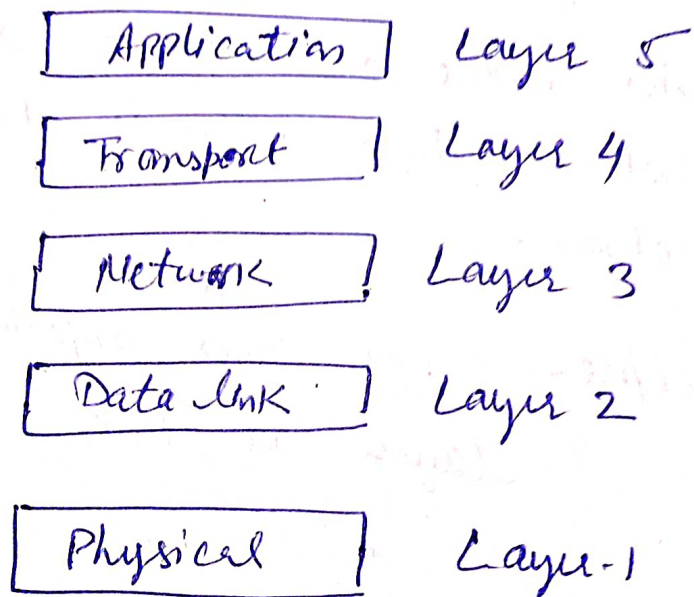
Logical Connections -

- layer to layer communication.

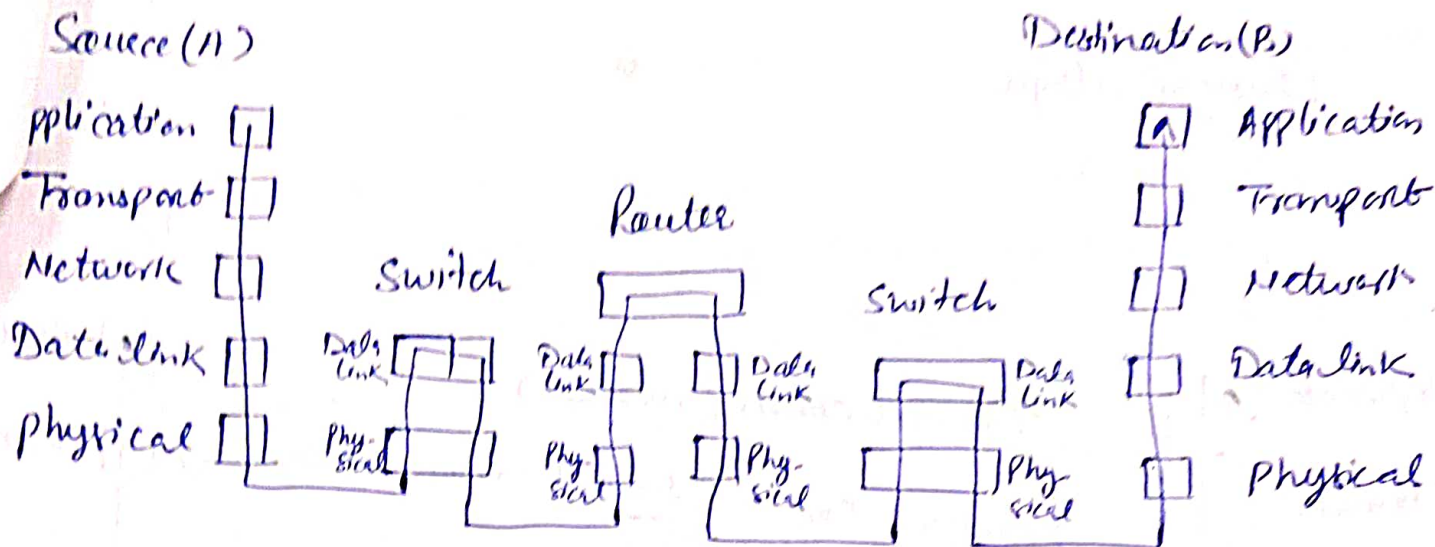


TCP/IP Protocol Suite → (Hierarchical Protocol
↓
(Transmission Control Protocol / Internet Protocol))

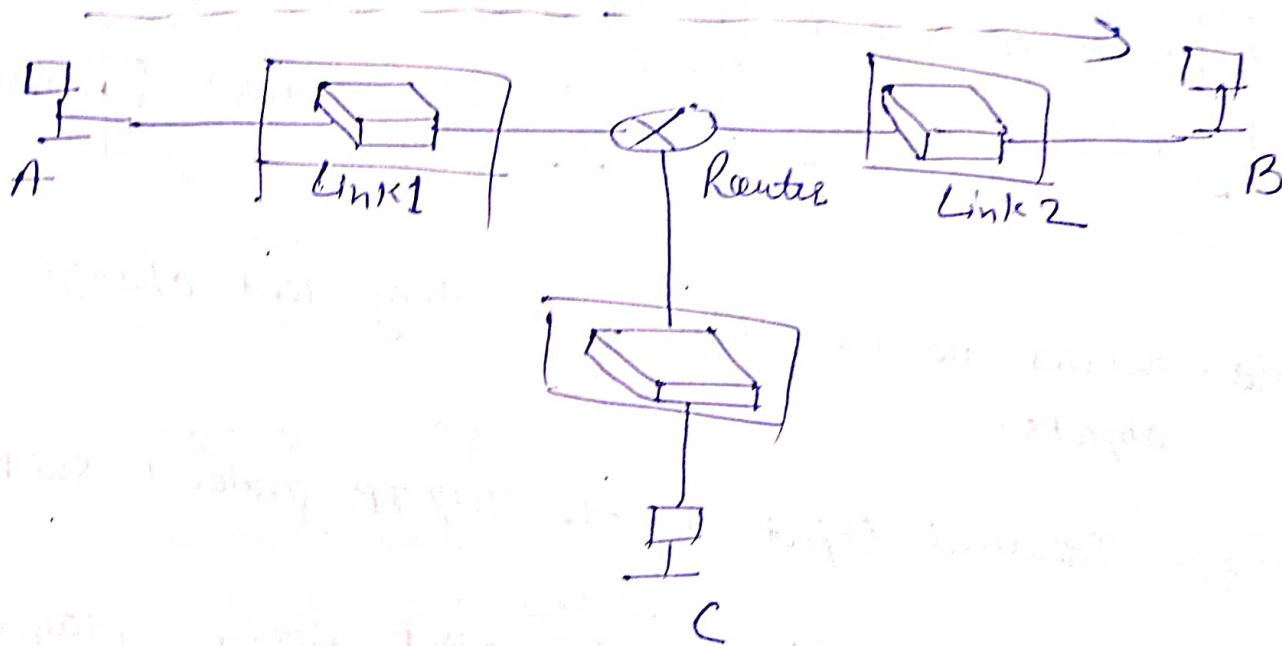
- TCP/IP is a protocol suite (a set of protocols) organized in different layers) used in the Internet today.
- TCP/IP protocol suit is defined as five layers.



Layered Architecture

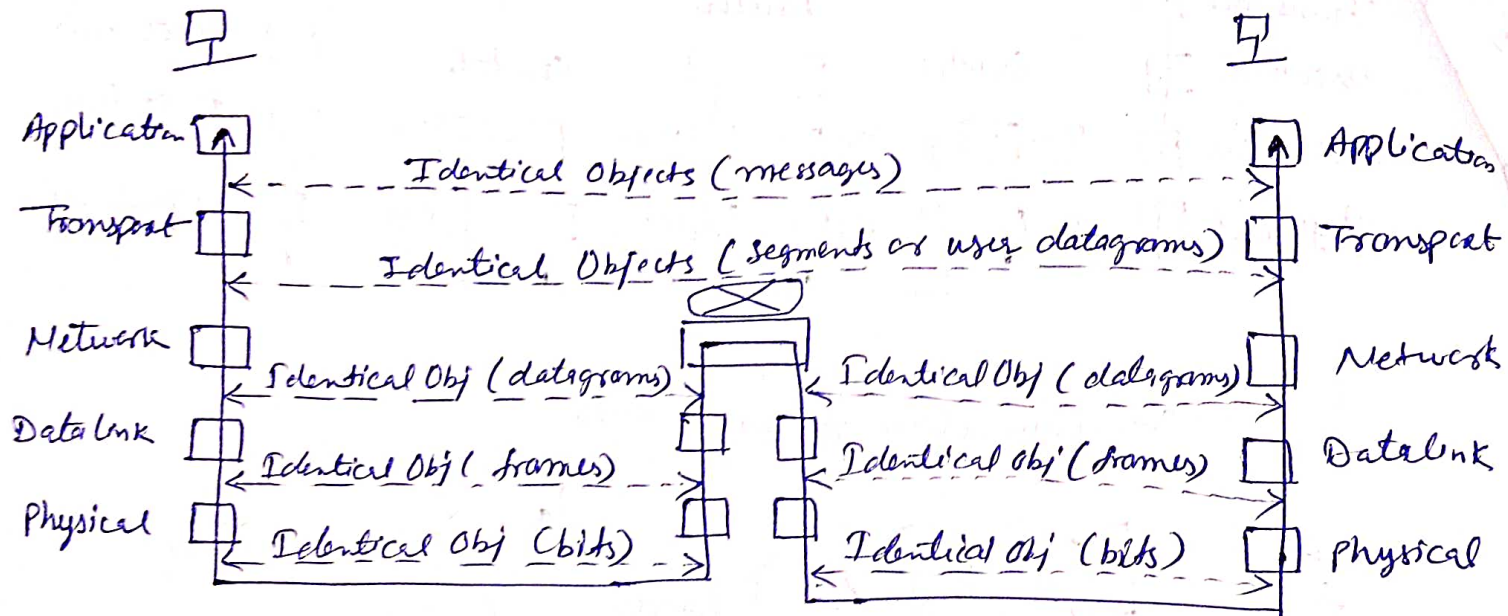


Communication from A to B



Description of each layer

Physical layer -



Note - switches are not shown as they don't change objects.

Fig:- Identical Objects in the TCP/IP protocol suit.

Logical connection → Application, Transport, Network layers.

end-to-end
hop-to-hop
Data Link }
Physical }
* hop is a host or router
domain of duty of these three layers is Internet.
and domain of duty of other two is link.

Another way of logical connection is shown in above fig.

- * data unit created from each layer. (packets) → should not be changed by router or link layer switch.
- In the Two layers, packet created by the host is changed only by the routers, not by the link layer switches.

Description of layers -

(12)

(11)

① - Physical layer -

- Responsible for carrying individual bits in a frame across the link.
- Communication b/w two devices at the physical layer is still a logical communication because there is another hidden layer, the transmission media, under the physical layer.

② - Data link layer -

- Internet is made up of several links (LANs & WANs) connected by routers.
- When the next link to travel is determined by the router, the data link layer is responsible for taking the datagram and moving it across the link.

③ - Network layer -

- responsible to create connection b/w source and destination.
- host-to-host communication
- because there can be several routes from the source to destination, the routers in the path are responsible for choosing the best route for each packet.

④ - Transport layer -

- end to end connection.
- Transport layer at source ~~end~~ host gets the message from the application layer, encapsulates it in a transport layer packet (called a segment or a user datagram in different protocols), and sends it ~~to~~ through the logical (imaginary) connection, to the transport layer at the destination host.
- Transport layer is responsible for giving services to the application layer.

⑤ - Application layer -

- end-to-end logical connection.
- exchange message b/w each other as through there were a bridge b/w the two layers.

The OSI Model

- Established in 1947, the International Organization for Standardization (ISO) is a multinational body dedicated to world wide agreement ^{international} on standards.
- Almost three fourths of the countries in the world are represented in the ISO.
- An ISO standard that covers all aspects of network communications is the "Open Systems Interconnection" (OSI) model.
- It was first introduced in the late 1970s.

ISO → is the organization
OSI → is the model.

- An Open System is a set of protocols that allows any two different systems to communicate regardless of their underlying architecture.
- The OSI model is not a protocol, it is a model for understanding and designing a n/w architecture that is flexible, robust and interoperable.
- OSI model is a layered framework for the designing of n/w s/s.
- It consists of seven separate but related layers.

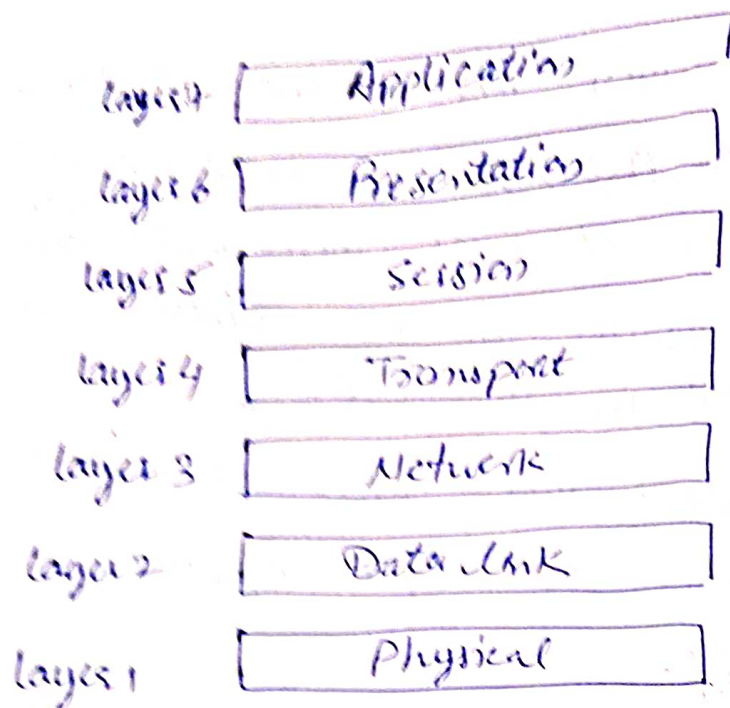
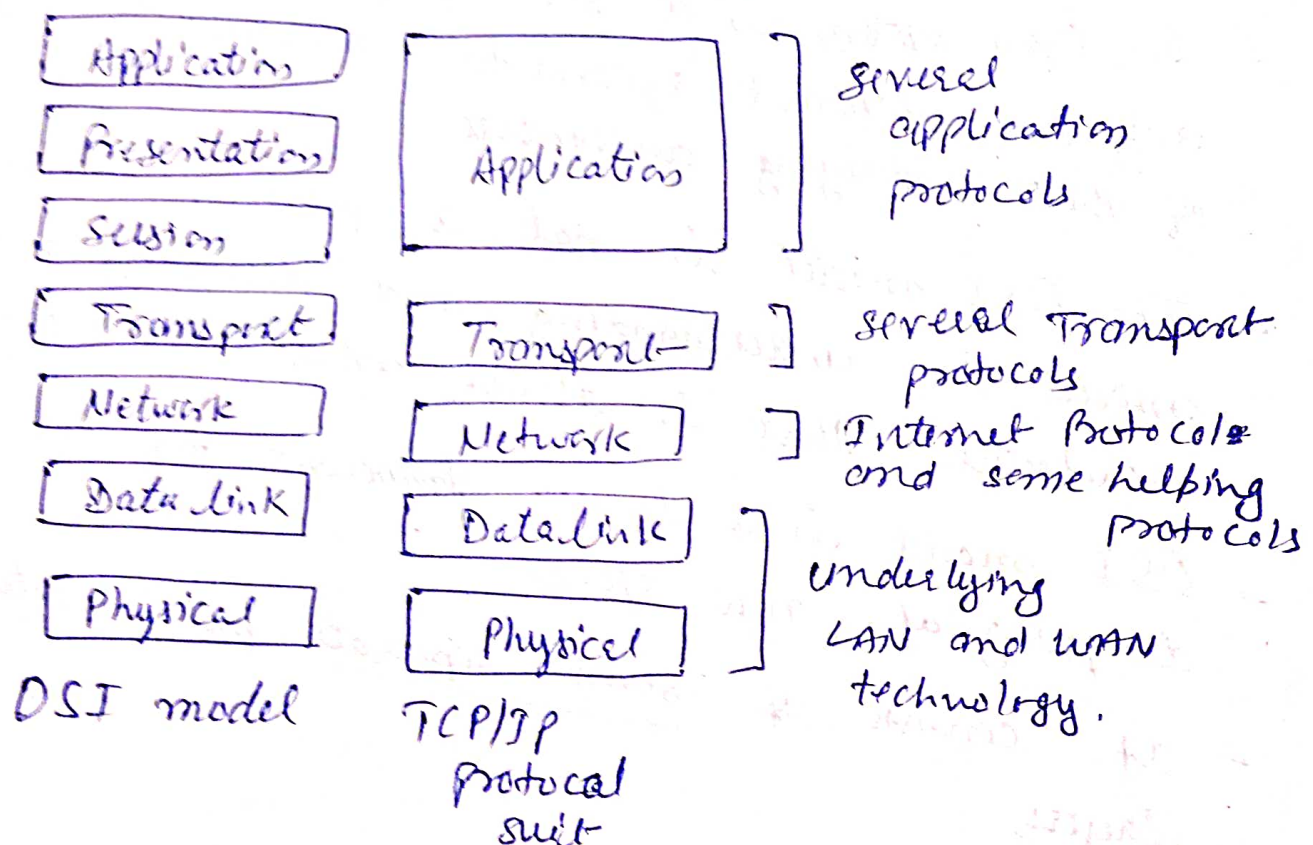


Fig - The OSI model

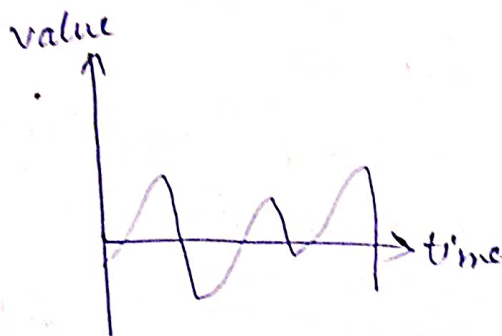
OSI versus TCP/IP



Physical Layer

(13)

Signals -



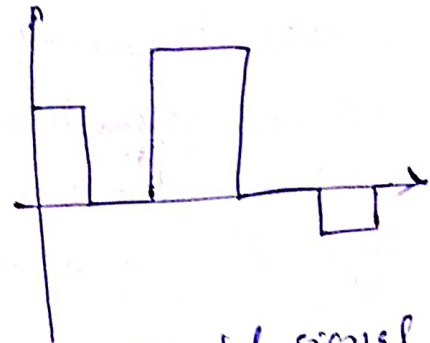
Analog Signal

- can be periodic or aperiodic
- In data comm. generally use periodic analog signals.

Parameters

- Period
- ~~Am~~ Peak Amp.
- phase
- frequency.
- wavelength. ($\lambda = c/f$)
- Bandwidth.

value



Digital signal

- takes limited no. of values.
- 1 $\rightarrow$ +ve
0 $\rightarrow$ -ve
- can have more than two levels.

parameters

- Bit rate
 $\rightarrow$ No. of bits sent/sec.
- Bit length
 $\rightarrow$ the distance 1 bit occupies on the transmission medium

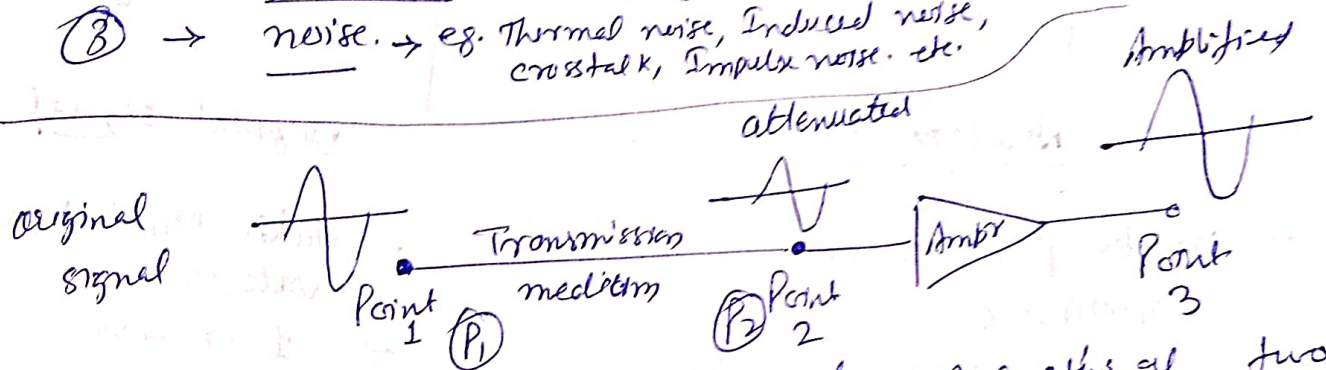
$$\text{Bit length} = \frac{1}{\text{Bit rate}}$$

Transmission of digital Signal

- (1) $\rightarrow$ Baseband Transmission $\rightarrow$ sending a digital signal over a channel w/o changing it to analog signal.
- (2) $\rightarrow$ Broadband Transmission
or
(modulation) $\rightarrow$ changing the digital signal to an analog signal for transmission.

Signal Impairment

- Signals travel through transmission media, where data are not perfect.
- Three causes of impairment are.
 - ① → attenuation → means loss of energy. (need amp^r to overcome it)
 - ② → distortion → changing its shape/form.
 - ③ → noise. → eg. Thermal noise, Induced noise, crosstalk, Impulse noise. etc.



→ Decibel (dB) measures the relative strengths of two signals or one signal at two different points.

$$\boxed{dB = 10 \log_{10} (P_2/P_1)}$$

Ex. A signal travels through transmission medium and its power is reduced to one-half.

$$\begin{aligned} \text{attenuation loss in (dB)} &= 10 \log_{10} \frac{P_2}{P_1} = 10 \log_{10} \frac{0.5P_1}{P_1} \\ &= -3 \text{ dB} \end{aligned}$$

Note- A loss of 3 dB (-3 dB) is equivalent to losing one half the power.

→ Signal to Noise Ratio (SNR).

$$\boxed{SNR = \frac{\text{avg signal P/W}}{\text{avg Noise P/W}}}$$

It is often described in dB.

Data rate limits

(14)

- Depends on three factors.

(1) - The bandwidth available

(2) - level of the signal we use.

(3) - quality of the channel (level of noise)

- Two theoretical formulas were developed to calculate the data rate -

By Nyquist $\rightarrow$ for a noiseless channel.

By Shannon $\rightarrow$ for a noisy channel.

$\rightarrow$ Nyquist bit rate defines the theoretical maximum bit rate.

$$\boxed{\text{Bit rate} = 2 \times B \times \log_2 L} \quad \text{no. of bits / second.}$$

$B \rightarrow$ Bandwidth.

$L \rightarrow$ no. of signal levels used to represent data.

Limitation - when we increase the number of signal levels, we impose a burden on the receiver.

~~For example~~ if no. of level is just 2, the receiver can easily distinguish b/w 0 and 1. But if level of signal is 64, then receiver must be very sophisticated to distinguish b/w 64 different levels.

$\rightarrow$ In other words, increasing the levels of a signal reduces the reliability of the s/s.

→ Noisy channel : Shannon Capacity -

- In reality we can't have noiseless channel.
- Claude Shannon introduced a formula, called the Shannon capacity, to determine the theoretical highest data rate for a noisy channel.

$$C = B \times \log_2 (1 + \text{SNR})$$

$B \rightarrow$ Bandwidth

$\text{SNR} \rightarrow$ signal to noise ratio

$C \rightarrow$ capacity of channel (bits/second)

Note $\rightarrow$ No signal levels.

i.e. we can't achieve a data rate higher than the capacity of the channel.

In other words, the formula defines the characteristics of the channel, not the method of transmission.

Example - we need a channel with a 1 MHz bandwidth. the SNR for this channel is 63. What are the approx. bit rate and signal level?

Solⁿ $\rightarrow$ Shannon formula to find the upper limit.

$$C = B \log_2 (1 + \text{SNR}) = 10^6 \log_2 (1 + 63) = 6 \text{ Mbps}$$

$\rightarrow$ for better performance we choose something lower, 4 Mbps.

Now use the Nyquist formula to find the no. of signal level,

$$4 \text{ Mbps} = 2 \times 1 \text{ MHz} \times \log_2 L \rightarrow \log_2 L = 2$$

$$L = 4$$

Performance -

(15)

* Bandwidth

→ in hertz → range of freq. involved.

→ in bits per seconds → no. of bits a channel can pass.

* Throughput

how fast we can actually send data through a network.

* Latency (Delay)

Defines how long it takes for an entire message to completely arrive at the destination from the time the first bit is sent out from the source.

- four types of delay -

- (i) - propagation delay
- (ii) - transmission delay
- (iii) - queuing delay
- (iv) - processing delay

$$\text{Latency} = \text{Prop. delay} + \text{Trans. delay} + \text{Que. delay} + \text{Proc. delay.}$$

* Bandwidth delay product

Defines the number of bits that can fill the link.

* Jitter (related to delay)

Digital Transmission -

- In digital Transmission, if data are digital, we need to use digital-to-digital conversion techniques.
- If data are analog, we need to use analog-to-digital conversion.

Digital to Digital Conversion -

The conversion involves three techniques -

- line coding
- block coding
- scrambling.

* Line coding is always needed.

* block coding and scrambling may or may not be needed.

Line Coding - process to convert the digital data into digital signals.

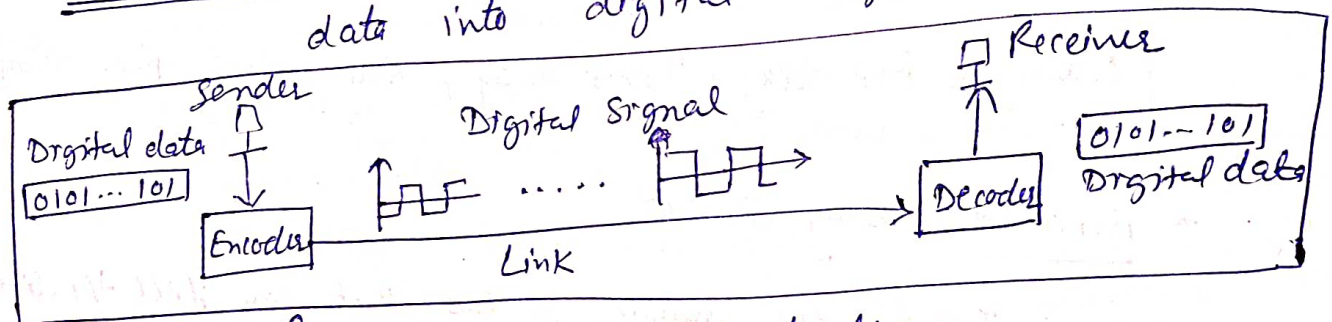


Fig- Line coding and decoding.

Block Coding - we need redundancy to ensure synchronization and to provide some kind of inherent error detecting.

- Block coding give this redundancy and improve the performance of line coding.

- In general, block coding changes a block of m bits into a number of n bits, where n is greater than m .

- Block coding normally involves three steps - division, substitution and combination.

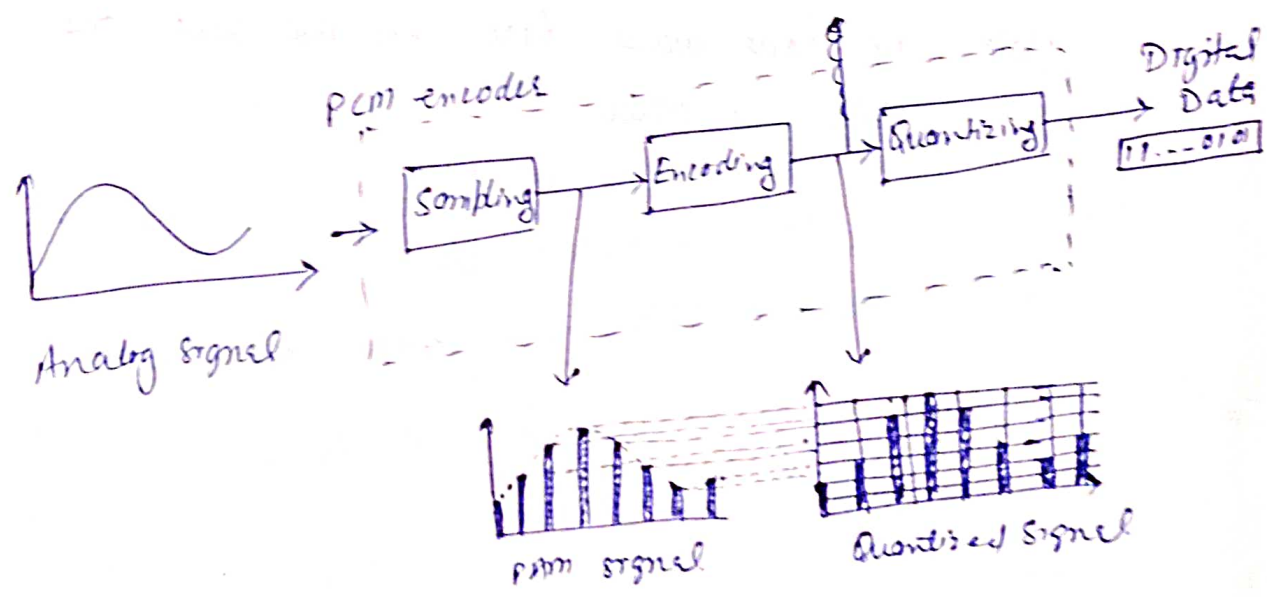
Analogy to digital Conversion -

→ Pulse code modulation

→ Delta modulation

PCM

- After the digital data are created (Digitization) it is converted to digital signal.

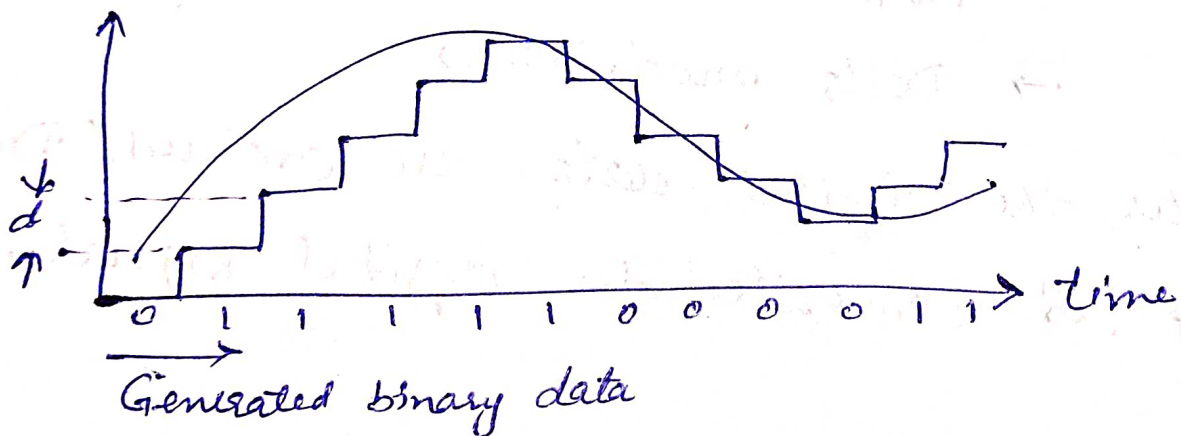


The three processes are -

- 1) - Analog signal is sampled every T_s .
- 2) - sampled signal is quantized, which means every sample is considered as pulse.
- 3) - The quantized value (pulses) are encoded as stream of bits.

Delta modulation -

- PCM is very complex.
- PCM finds the value of the signal amplitude for each sample.
- DM finds the change from the previous sample.



Note - No code word here, bits are sent one after the other.

Analog Transmission -

(17) @

- In digital transmission - it needs a low-pass channel (a channel that starts from 0)
- Analog transmission is the only choice for band pass channel (a channel that does not start from 0).
- Converting digital data to a bandpass analog signal → Digital to analog conversion.
- Converting a low-pass analog signal to bandpass signal → Analog to analog conversion.

Digital to analog conversion -

- ASK
- FSK
- PSK
- Fourth (and better) mechanism that combines changing both the amplitude and phase, called Quadrature Amplitude Modulation (QAM)

Analog to Analog conversion -

- AM
- FM
- PM

Multiplexing -

- FDM → analog Technique.
- TDM → Digital Process
- * WDM → wavelength division multiplexing is also analog Tech.

Transmission Media

→ Transmission media are actually located below the physical layer and are directly controlled by physical layer.

→ Transmission media can be defined as anything that can carry information from a source to a destination.

→ In data communication -

Transmission media are - Free space, metallic cable, fiber-optic cable.

Guided Media -

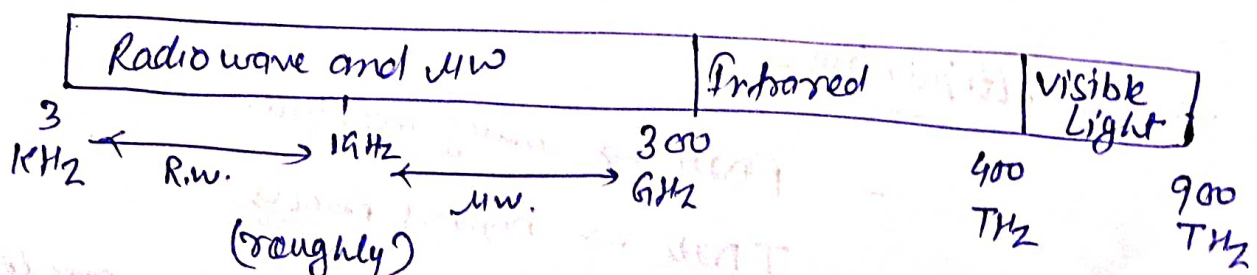
metallic (copper) $\left[\begin{array}{l} \rightarrow \text{Twisted pair cable} \\ \rightarrow \text{coaxial cable} \\ \rightarrow \text{Fiber-optic cable} \end{array} \right. \begin{array}{l} \rightarrow \text{Unshielded (UTP)} \\ \rightarrow \text{Shielded (STP)} \end{array}$

Unguided Media -

→ Transport EM waves w/o using a physical conductor.

- This is called wireless communication.

- EM spectrum ranging from 3 kHz to 900 THz.



Radio frequency spectrum -

ELF - 3-30 kHz

SLF - 30-300 kHz

ULF - 300-3 kHz

VLF - 3 kHz - 30 kHz

LF - 30 kHz - 300 kHz

MF - 300 kHz - 3 MHz

HF - 3 MHz - 30 MHz

VHF - 30 MHz - 300 MHz

UHF - 300 MHz - 3 GHz

SHF - 3 GHz - 30 GHz

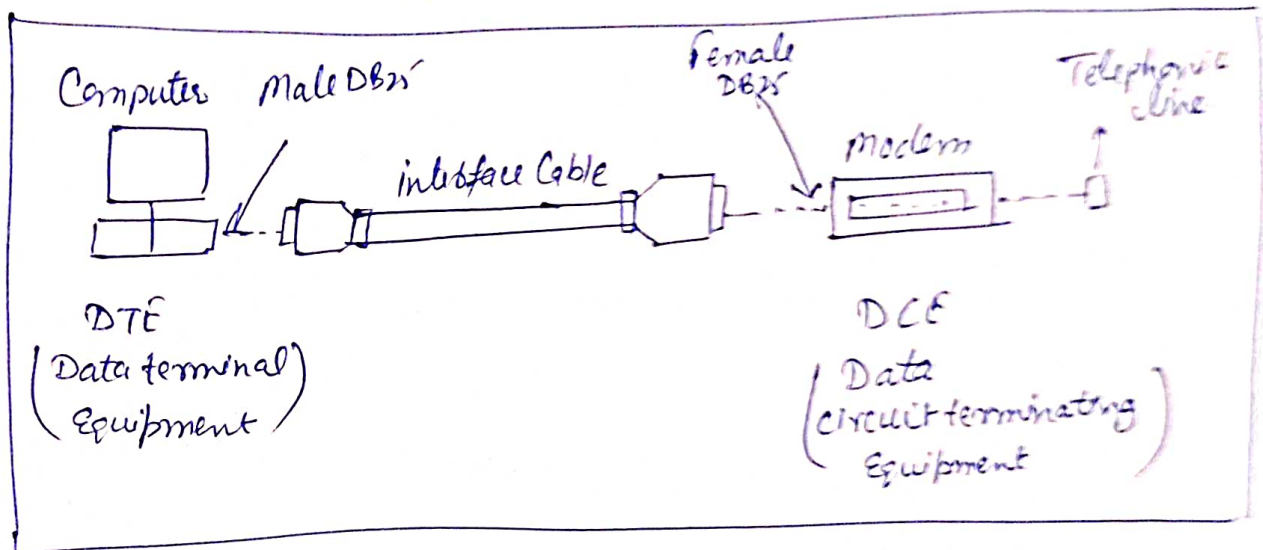
EHF - 30 GHz - 300 GHz

DTE- DCE Interface

DTE → Data termination Equipment

DCE → Data Communication Equipment / Data Circuit Terminating Equipment

- It is a device that is an information source or an information sink.
- It is a device used as an interface between a DTE.



- DTE is any device that is a source of or destination for binary digital data.
- at the physical layer, it can be terminal, micro-computer, computer, printer, fax machine.
- DTEs do not often communicate directly with one another.
- A DCE is any device that transmits or receives data in the form of an analog or digital signal through a network.
- At the physical layer, DCE takes a data generated by a DTE.
- Commonly used DCE is modem.

Interface standards

It is a standard that describes one or more functional characteristics (such as code conversion, line assignments, or protocol compliance) or physical characteristics (as electrical, mechanical, or optical) necessary to allow the exchange of information b/w two or more systems.

Modems - (modulator - demodulator)

A device that accepts a serial stream of bits as input and produces a modulated carrier as output (or vice versa) is called a modem.

- The modem is inserted b/w the (digital) computer and the (analog) telephone system.
- To go to higher speed, it is not possible to increase the sampling rate.
- Nyquist says $\rightarrow$ with a perfect 3000 Hz line, there is no point in sampling faster than 6000 Hz.
- Thus all research on faster modems is focused on getting more bits per sample (i.e. per baud).